

JITHIN K. SREEDHARAN

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Seattle
Washington
U.S.A.

Professional Summary

- Machine Learning (ML) Researcher and Data Scientist with a strong background in statistical modeling, data science, and machine learning, and experience in applying data science techniques to a wide range of domains, including e-commerce, health and social sector, computational social science, bioinformatics, and network science
- Experience in multiple stages of ML solutions — data procurement, feature engineering, algorithm development, and offline and online evaluation
- Strong applied and theoretical research record leading to publications in selective venues including ICML, SIGKDD, The Web Conference (WWW), SIGMETRICS, Nature Reports, INFOCOM, etc.

Employment

The Home Depot

Seattle, WA, USA

Senior Data Scientist

10/2022 – present

- Part of the core Recommendation Systems ML team and sole module owner of the add-to-cart overlay recommendations
- Spearheaded learning-to-rank based systems inside the recommendation team – classical, deep learning, and LLM based techniques.
- Developed and maintained a personalized ranking solution that brought multi-million-dollar in annualized incremental revenue in online A/B test
- Developed a graph neural network based recommendation retrieval system that performs better than existing cross-selling retrieval modules in the team
- Mentored, developed, and fine-tuned a Small Language Model based ranker
- Runner up of The Home Depot Data Science Hackathon 2022

Indian Institute of Technology Kanpur

Kanpur, India

Dept. of Computer Science and Engineering

Assistant Professor (full time; permanent)

04/2022 – 01/2023

- Research on machine learning (especially that on graphs), A.I. for Social Good, and general graph mining
- Guided one MS (thesis) student and a summer intern

Wadhvani Institute for Artificial Intelligence: AI for Social Impact

Machine Learning Scientist and Research Manager

Mumbai, India · 01/2022 – 04/2022

Los Altos, CA, USA · 10/2020 – 12/2021

- Developed AI for Social Good and Health solutions to help improve the lives of millions. Our collaborators and funders included **Google.org**, **Bill & Melinda Gates Foundation**, United States Agency for International Development, Harvard University, WHO, and International Innovation Corps (UChicago)
- Led two teams of Associate ML Scientists and coordinated with engineering and product teams, on a challenging behavioral prediction problem and a cough-based triaging solution for Tuberculous infected patients. Both the solutions are deployed country-wide in India, and have a huge social impact on Indian healthcare system. Our paper on the behavior prediction project bagged **the outstanding paper award in ML4H** (ML for Health) NeurIPS Symposium, 2022
- As of December 2024, the behavioral prediction ML solution has served 120,000 patients across 10 states in India. Starting January 2025, it is projected to serve approximately 2.4 million patients annually

National Science Foundation (NSF) Center for Science of Information

Department of Computer Science, Purdue University

West Lafayette, IN, USA

NSF Postdoctoral Researcher

01/2017 – 10/2020

Visiting Research Scholar

04/2016 – 06/2016

Mentors: Prof. Wojciech Szpankowski and Prof. Ananth Grama

- Developed an optimal feature selection algorithm for explainable AI solutions and various approximate solutions for it with reduced time and sample complexity
- Designed and executed research on unsupervised and semi-supervised techniques for graph mining and embedding – recovering temporal information hidden in the dynamic data of social media and biological systems
- Coordinated a team consisting of three senior professors and three postdoctoral researchers, and instituted research collaboration with an interdisciplinary data science team in quantum computing
- Resulted in publications (6 conferences incl. **ICML**, **The Web Conference (WWW)** and **SIGKDD**, 3 journal incl. Nature), 4 grant proposals, and 3 open source libraries. Delivered 17 invited research talks (incl. Google Research, Adobe Research, Bell Labs, TU Eindhoven, SUTD Singapore, IISc Bangalore, IIT - Hyderabad, and IIT Bombay, Delhi, Madras, and Kanpur)

Institut National de Recherche en Informatique et en Automatique (INRIA) and

INRIA-Bell Labs joint lab

Sophia Antipolis and Paris, France

PhD Graduate Researcher, Team MAESTRO (renamed to NEO)

08/2013 – 12/2016

Advisor: Dr. Konstantin Avrachenkov

- Designed and analyzed distributed unsupervised data mining algorithms for graphs to sample, rank and estimate graph properties.
- Developed methods based on reinforcement learning, short random walks, extreme value theory, and spectral graph theory for estimation problems on networked data
- Designed distributed implementation of spectral clustering techniques
- Launched collaborations with researchers from Purdue/CMU (USA), Bell Labs, UFRJ (Brazil), IIT Bombay (India), and IISc Bangalore (India)
- Resulted in publications (5 conferences, 3 journal), and 4 open source libraries. Delivered 3 invited and 4 contributed research talks

Indian Institute of Science

Bangalore, India

Performance Analysis Lab, Dept. of Electrical Communication Engineering

Research Associate

09/2010 – 04/2013

Advisor: Prof. Vinod Sharma

- Developed and analyzed sequential hypothesis testing algorithms for distributed quickest detection of data anomalies, with various generalizations from parametric to non-parametric setup
- Tested the devised methods in anomaly detection in wireless sensor networks and spectrum sensing in cognitive radios, with aid from Boeing Inc. and Ministry of Communications and Information Technology, Govt. of India
- Resulted in publications (5 conferences, 1 journal), and 2 open source libraries

Robert Bosch

Coimbatore and Bangalore, India

System Engineer, Automotive embedded systems in gasoline engines

08/2007 - 12/2008

- Developed and maintained software for engine control units (ECUs)
- Integrated new device drivers and application software modules into a common platform after rigorous hardware and software testing.

Education

Doctor of Philosophy in Computer Science

08/2013 – 12/2016

Institut National de Recherche en Informatique et en Automatique (INRIA) and

INRIA-Bell Labs joint lab

Sophia Antipolis and Paris, France

Affiliated university: Université Côte d’Azur (Université Nice Sophia Antipolis), France

Thesis title: *Sampling and Inference in Complex Networks*

Master of Science (Engineering), Dept. of Electrical Communication Engg. 09/2009 – 07/2012
Indian Institute of Science (IISc) Bangalore, India

Thesis title: *Spectrum Sensing in Cognitive Radios using Distributed Sequential Detection*

Bachelor of Technology in Electronics and Communication Engineering 08/2003 – 05/2007
Govt. Model Engineering College - Cochin University of Science and Technology Cochin, India

Main project title: *FPGA Implementation of a Probabilistic Neural Network*

Fellowships, Awards, and Honors

- Outstanding paper award at NeurIPS 2022 Machine Learning for Health (ML4H) Symposium
- Postdoctoral fellowship from NSF Science and Technology Center for Science of Information
- Offered the full time position of Assistant Professor in International Institute of Information Technology Hyderabad (IIIT-H), Hyderabad, India (Declined)
- Postdoctoral fellowship from University of California San Diego Halicioglu Data Science Institute (Declined)
- ACM SIGMETRICS/PERFORMANCE travel grant 2016
- **Best M.S. thesis medal** - Prof. F. M. Mowdawalla medal - from Indian Institute of Science, Bangalore, India
- PhD fellowship from INRIA-Bell Labs (France) joint lab for the entire duration of PhD
- Ministry of Human Resources and Development (MHRD), Govt. of India, scholarship for graduate studies
- Offered Scientist-C position at Defence Research and Development Organisation (DRDO), a prestigious public defence research lab under Govt. of India (Declined)
- Secured a position in top 0.1 percentage of candidates (from nearly 100,000) in the admission test (GATE 2009, Electronics and Communication paper) for graduate studies in India

Publications

- *In all the papers, I am either one of the two main contributors or the main contributor.*
- *The asterisk (*) symbol indicates that the paper follows an alphabetical author-list*
- 1. **Predicting Treatment Adherence of Tuberculosis Patients at Scale**
with Wadhvani AI, USAID, Central TB Division (India), and WHO teams
NeurIPS Machine Learning for Health (ML4H) Symposium, Proc. of Machine Learning Research, 2022
<https://proceedings.mlr.press/v193/kulkarni22a> [Symposium proceedings]
Outstanding paper award, I led the ML/Data Science team on this project.
- 2. **Deploying Covid-19 Case Forecasting Models in the Developing World**
with Wadhvani AI team
Chapter in the book - *AI for Social Impact*, 2022 [Book chapter]
- 3. **Sufficiently Informative and Relevant Features: An Information-theoretic and Fourier-based Characterization**
Mohsen Heidari, Jithin K. Sreedharan, Gil Shamir, and Wojciech Szpankowski
IEEE Transactions on Information Theory, 2022
doi:10.1109/TIT.2022.3169998 [Journal paper]
- 4. **Finding Relevant Information via a Discrete Fourier Expansion**
Mohsen Heidari, Jithin K. Sreedharan, Gil Shamir, and Wojciech Szpankowski
International Conference on Machine Learning (ICML), 2021
Acceptance rate: 18.46%; supplementary Material
<http://proceedings.mlr.press/v139/heidari21a.html> [Conference proceedings]

5. [Information Sufficiency via Fourier Expansion](#)
Mohsen Heidari, *Jithin K. Sreedharan*, Gil Shamir, and Wojciech Szpankowski
IEEE International Symposium on Information Theory (ISIT), 2021 [Conference proceedings]
[doi:10.1109/ISIT45174.2021.9517924](#)
6. [Temporal Ordered Clustering in Dynamic Networks: Unsupervised and Semi-Supervised Learning Algorithms](#)
Krzysztof Turowski[†], *Jithin K. Sreedharan*[†], and Wojciech Szpankowski; [†]*Equal contribution*
(a) *IEEE International Symposium on Information Theory (ISIT)*, 2020
[doi:10.1109/ISIT44484.2020.9174079](#) [Conference proceedings]
(b) *IEEE Transactions on Network Science and Engineering*, vol. 8, no. 2, pages 1426-1442, 2021
[doi:10.1109/TNSE.2021.3058376](#) [Journal paper]
7. [Revisiting Parameter Estimation in Biological Networks: Influence of Symmetries](#)
Jithin K. Sreedharan[†], Krzysztof Turowski[†], and Wojciech Szpankowski; [†]*Equal contribution*
(a) *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2019 (poster presentation)
(b) *BioKDD 2019* – in conjunction with SIGKDD (oral presentation)
https://biokdd.org/biokdd19/regular_track.html [Workshop proceedings]
(c) *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2020
[doi:10.1109/TCBB.2020.2980260](#) [Journal paper]
8. [Inferring Temporal Information from a Snapshot of a Dynamic Network](#)
Jithin K. Sreedharan[†], Abram Magner[†], Ananth Grama, and Wojciech Szpankowski; [†]*Equal contribution*
Nature Scientific Reports, 10 pages (and 20 pages supplementary material), 2019
Technical details with theoretical analysis and proofs are in the *supplementary material*
[doi:10.1038/s41598-019-38912-0](#) [Journal paper]
9. [TIMES: Temporal Information Maximally Extracted from Structures](#)
Abram Magner[†], *Jithin K. Sreedharan*[†], Ananth Grama, and Wojciech Szpankowski; [†]*Equal contribution*
World Wide Web Conference (WWW), 2018
Acceptance rate: 14.8%; selected for oral presentation in the main conference
[doi:10.1145/3178876.3186105](#) [Conference proceedings]
- *10. [Revisiting Random Walk based Sampling in Networks: Evasion of Burn-in Period and Frequent Regenerations](#)
Konstantin Avrachenkov, Vivek S. Borkar, Arun Kadavankandy and *Jithin K. Sreedharan*
Computational Social Networks (Springer International Publishing), 5(1), 19 pages, 2018
[doi:10.1186/s40649-018-0051-0](#) [Journal paper]
- *11. [Hamiltonian System Approach to Eigenvalue-Eigenvector Problem in Networks](#)
Konstantin Avrachenkov, Philippe Jacquet and *Jithin K. Sreedharan*
IEEE International Workshop on Multidimensional (nD) Systems (nDS), 2017
[doi:10.1109/NDS.2017.8070631](#) [Workshop proceedings]
12. [Recovery of Vertex Orderings in Dynamic Graphs](#)
Abram Magner, Ananth Grama, *Jithin K. Sreedharan* and Wojciech Szpankowski
IEEE International Symposium on Information Theory (ISIT), 2017
[doi:10.1109/ISIT.2017.8006792](#) [Conference proceedings]
- *13. [Inference in OSNs via Lightweight Partial Crawls](#)
Konstantin Avrachenkov, Bruno Ribeiro and *Jithin K. Sreedharan*
(a) *SIGMETRICS Performance Evaluation Review*, 44(1), pages 165-177, 2016
[doi:10.1145/2964791.2901477](#) [Journal paper]
(b) *ACM SIGMETRICS/IFIP International Conference on Measurement and Modeling of Computer Science (SIGMETRICS '16)*, 2016

Acceptance rate: 13.5%; selected for oral presentation in the main conference.
[doi:10.1145/2896377.2901477](https://doi.org/10.1145/2896377.2901477) [Conference proceedings]

- *14. [Distributed Spectral Decomposition in Networks by Complex Diffusion and Quantum Random Walk](#)
Konstantin Avrachenkov, Philippe Jacquet and *Jithin K. Sreedharan*
IEEE International Conference on Computer Communication (INFOCOM'16), 2016
Acceptance rate: 18.25%; selected for oral presentation in the main conference.
[doi:10.1109/INFOCOM.2016.7524376](https://doi.org/10.1109/INFOCOM.2016.7524376) [Conference proceedings]
- *15. [Comparison of Random Walk Based Techniques for Estimating Network Averages](#)
Konstantin Avrachenkov, Vivek S. Borkar, Arun Kadavankandy and *Jithin K. Sreedharan*
Springer LNCS, 5th International Conference on Computational Social Networks (CSoNet), vol 9795, pages 27-38, 2016
[doi:10.1007/978-3-319-42345-6_3](https://doi.org/10.1007/978-3-319-42345-6_3) [Conference proceedings]
- *16. [Distribution and Dependence of Extremes in Network Sampling Processes](#)
Konstantin Avrachenkov, Natalia M. Markovich and *Jithin K. Sreedharan*
Computational Social Networks (Springer International Publishing), 2(1), 21 pages, 2015
[doi:10.1186/s40649-015-0018-3](https://doi.org/10.1186/s40649-015-0018-3) [Journal paper]
- 17. [Spectrum Sensing using Distributed Sequential Detection via Noisy Reporting MAC](#)
Jithin K. Sreedharan and Vinod Sharma
Signal Processing (Elsevier & EURASIP), vol 106, pages 159-173, 2015
[doi:10.1016/j.sigpro.2014.07.009](https://doi.org/10.1016/j.sigpro.2014.07.009) [Journal paper]
- 18. [Nonparametric Distributed Sequential Detection via Universal Source Coding](#)
Jithin K. Sreedharan and Vinod Sharma
IEEE Information Theory and Applications Workshop (ITA), 2013
[doi:10.1109/ITA.2013.6502977](https://doi.org/10.1109/ITA.2013.6502977) [Workshop proceedings]
- 19. [Spectrum Sensing via Universal Source Coding](#)
Jithin K. Sreedharan and Vinod Sharma
IEEE Global Communications Conference (GLOBECOM), 2012
[doi:10.1109/GLOCOM.2012.6503327](https://doi.org/10.1109/GLOCOM.2012.6503327) [Conference proceedings]
- 20. [Novel Algorithms for Distributed Sequential Hypothesis Testing](#)
K. S. Jithin and Vinod Sharma
IEEE Annual Allerton Conference on Communication, Control, and Computing, 2011
[doi:10.1109/Allerton.2011.6120349](https://doi.org/10.1109/Allerton.2011.6120349) [Conference proceedings]
- 21. [A Novel Algorithm for Cooperative Distributed Sequential Spectrum Sensing in Cognitive Radio](#)
Jithin K. Sreedharan and Vinod Sharma
IEEE Wireless Communications and Networking Conference (WCNC), 2011
[doi:10.1109/WCNC.2011.5779420](https://doi.org/10.1109/WCNC.2011.5779420) [Conference proceedings]
- 22. [Cooperative Distributed Sequential Spectrum Sensing](#)
K. S. Jithin, Vinod Sharma, and Raghav Gopalathnam
IEEE National Conference on Communication (NCC), 2011
[doi:10.1109/NCC.2011.5734763](https://doi.org/10.1109/NCC.2011.5734763) [Conference proceedings]

Community Service

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- Reviewer: ICDM, NeurIPS, SIGMETRICS, KDD, ISIT, ISMB/ECCB, Performance Evaluation, ACM Transactions on Modeling and Performance Evaluation of Computing Systems, IEEE Transactions on Information Theory, Algorithmica, ACM/IEEE Transactions on Networking, IEEE Transactions on Systems, Man, and Cybernetics: Systems, Physica A, IEEE Transactions on Network Science and Engineering
 - Red Cross Disaster Action Team (DAT) on-call trained volunteer in Seattle, US.